RV College of Engineering®, Bengaluru – 59

Department of Artificial Intelligence and Machine Learning

Database Management Systems (CD252IA)

Synopsis

TITLE: Sentinel AI: AI-Based Remote Server Monitoring and Alerting Database System		
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Introduction:

Sentinel AI is an AI-based remote server monitoring and alerting system designed to continuously track, store, and analyze real-time system metrics from distributed servers. The system monitors critical performance parameters such as CPU utilization, memory usage, disk usage, temperature, and system activity, which are collected by a lightweight Python-based monitoring agent.

These metrics are transmitted to a central Flask backend using HTTP APIs and stored in a relational database (RDBMS). A modern React-based dashboard visualizes server health, performance trends, and alerts in real time.

To enhance reliability and automation, Sentinel AI integrates an unsupervised machine learning model (Isolation Forest) that analyzes historical monitoring data to detect anomalous behavior such as abnormal CPU spikes, memory leaks, or overheating. The system generates early alerts before failures occur, enabling proactive server management.

This project combines DBMS concepts, backend APIs, frontend visualization, and AI-driven analytics into a unified intelligent monitoring platform.

Existing System

Conventional server monitoring systems suffer from:

- Heavy manual dependency
- Lack of automation in failure detection
- No intelligence for anomaly detection
- No prediction of future overload
- Limited database integration
- Poor visualization of long-term trends

Proposed System:

The proposed Sentinel AI system introduces an AI-enabled, database-driven monitoring architecture that provides intelligent, automated, and proactive server monitoring.

Key Highlights:

- Automated data collection using monitoring agents
- Centralized relational database storage
- AI-based anomaly detection using Isolation Forest
- Early alert generation
- Interactive web dashboard for visualization

Difference from existing system:

Traditional Monitoring	Proposed AI-Based System
Static thresholds	AI-based prediction
Manual diagnosis	Automatic detection
No forecasting	Failure prediction
Pure monitoring	Smart monitoring
No intelligence	AI-driven insights

Relational Database Structure

The system uses a structured RDBMS (MySQL) to efficiently store and manage server monitoring data. The main tables include **Servers, Metrics, Logs, Thresholds, Alerts, and Users**, which ensure proper data organization and relationship management using primary and foreign keys.

An additional table called **AI_Predictions** is used to store AI-generated insights:

- anomaly_id (PK)
- server_id (FK)
- anomaly_type
- severity
- detected_time

AI + RDBMS + MONITORING INTEGRATION

- Monitoring Layer: Python agent collects system metrics
- Backend Layer: Flask APIs receive and process data
- Database Layer: Stores structured monitoring data
- AI Layer: Isolation Forest analyzes historical data
- Visualization Layer: React dashboard displays insights

Societal Concern

The system enhances reliability by detecting failures early and reducing manual monitoring efforts. It helps prevent downtime in critical sectors such as **banks, hospitals, and IT offices** while lowering maintenance costs and improving operational efficiency.

Technology Trends Used

- IoT
- AI-driven analytics
- Cloud databases
- Time-series forecasting
- Automation and predictive maintenance